

A Middle Eastern Government Partners with Torry Harris to Transform Car Rental Oversight

Onboards 2,888 Agencies and
Processes Over 4.19 Million
Contracts

www.torryharris.com/ae-en





Who

A government entity in a Middle Eastern country

What

The objective was to modernize and digitize the regulation and tracking of car rental operations across the country. The organization aimed to eliminate manual inefficiencies, improve transparency, and build a scalable, secure platform capable of integrating with other government systems and supporting future growth.

How

Torry Harris Integration Solutions (THIS) implemented a comprehensive digital platform using blockchain for secure record-keeping, containerized microservices for scalability, and an API Gateway for integration and third-party access. The system architecture emphasized high availability, performance, and cost-efficiency through Kubernetes and DevOps-based deployment.

Results

Widespread adoption: 2,888 car rental agencies have been onboarded. 128,084 local and 41,984 regional vehicles are now integrated into the system.

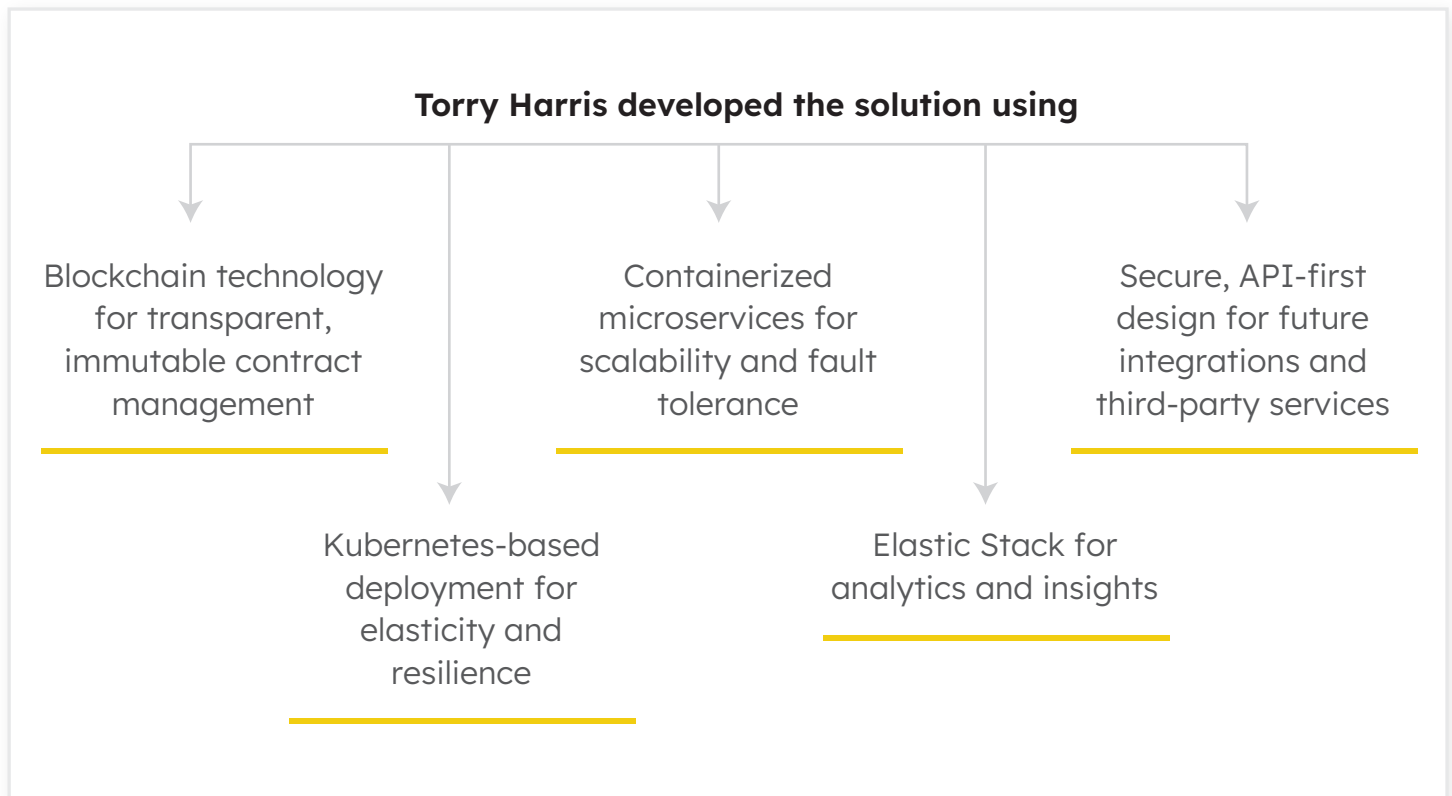
High transaction volume: Over 4.19 million rental contracts processed, and more than 520,000 transactions handled.

Revenue generation: APIs created new monetization opportunities, while digital automation improved service offerings and attracted more participants.

A scalable, transparent solution powered by blockchain and cloud-native design

The organization faced challenges including manual paperwork, limited transparency, fraudulent practices, and a lack of centralized oversight. To solve these issues, a digital solution was designed and deployed to provide real-time visibility, automate processes, and create a tamper-proof system for contract handling.

THIS was entrusted with complete ownership of a comprehensive digital platform's architecture, development, and implementation. The initiative delivered a robust end-to-end solution, tailored to address critical business requirements. Key components of the project included the deployment of a blockchain-based distributed ledger to ensure secure and tamper-proof record-keeping, and the creation of an API Gateway to facilitate seamless external integrations and efficient platform exposure.



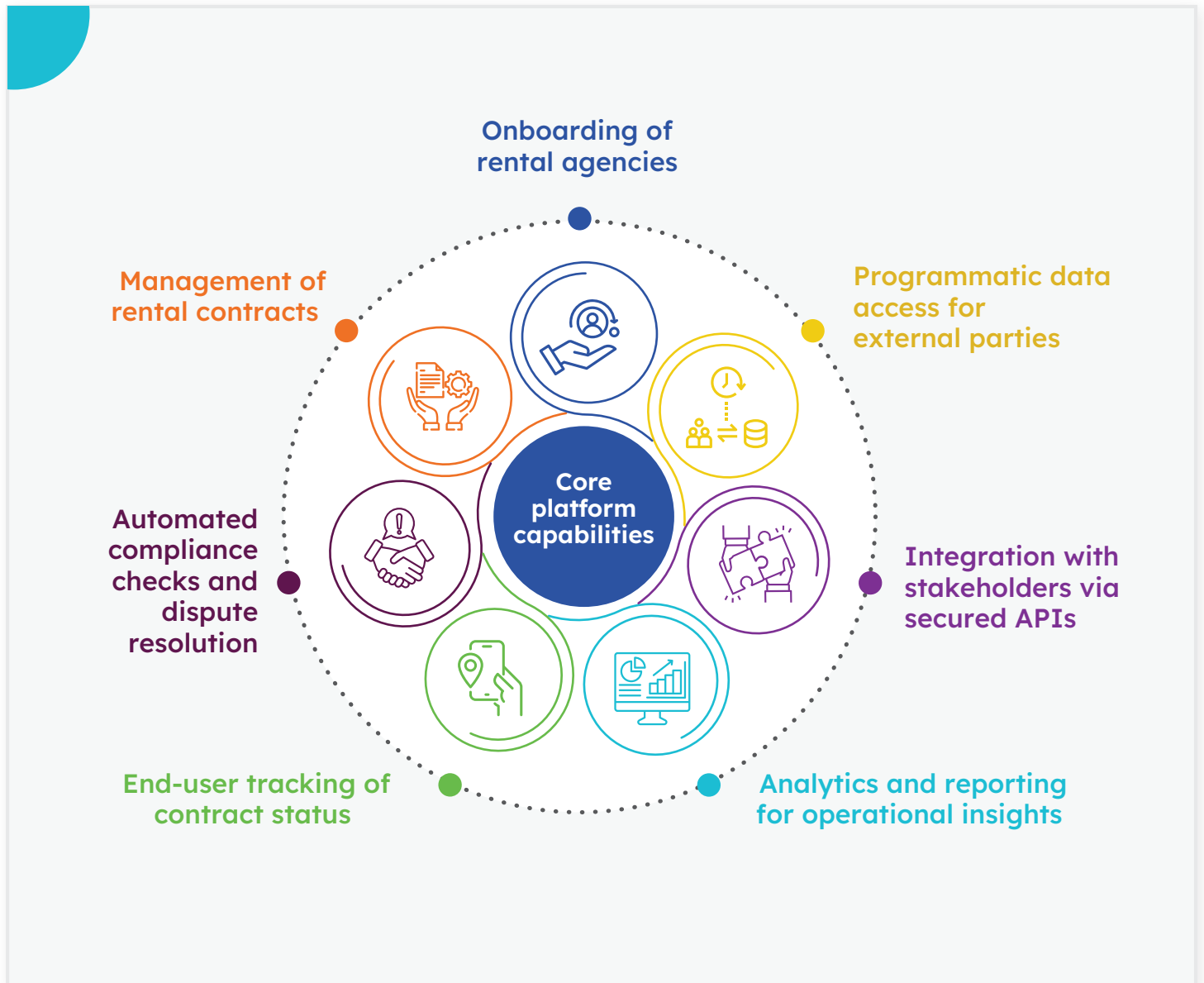
The platform maintains 100% availability and includes Level 2/3 managed services to ensure uninterrupted operation. The strategic implementation of IBM API Connect served as the main gateway for all service integrations, boosting operational efficiency.

THIS also built microservices for master data management, workflow management, and served as a wrapper for external integrations, ensuring smooth and efficient operations. The deployment model was designed to be containerized and scalable, utilizing Kubernetes and OpenShift to support future growth. Additionally, THIS developed microsites and integrated powerful analytics capabilities through the Elastic Stack, enhancing data-driven insights. To ensure ongoing reliability and performance, THIS provided managed services, including Level 2 and Level 3 support, guaranteeing the platform's continuous availability and stability.

By addressing key pain points, Torry Harris helped create a more efficient, secure, and scalable car rental management system that supported the organization's strategic objectives and fostered growth within the sector. The use of containerized deployment with Kubernetes and DevOps practices ensured scalability, reliability, and cost-efficiency, while maintaining 100% system availability. Additionally, the integration of Blockchain technology for issuing smart car rental contracts marked a groundbreaking milestone in the digital transformation strategy, making it the first of its kind in the region.

THIS builds secure, scalable platform with blockchain and real-time data insights

Through collaborative planning workshops, business and technical requirements were defined to ensure alignment with strategic goals. The result was a comprehensive platform, not just an application, designed for extensibility and ease of maintenance.



THIS collaborated closely with the organization stakeholders through workshops to define both functional and non-functional requirements, ensuring the solution aligned with business objectives. Based on these discussions, the project's architecture and technology stack were finalized.

- **Hyperledger Fabric**
for secure, tamper-proof smart contracts
- **IBM API Connect with OpenShift**
for exposing and managing APIs
- **Vue.js, Nuxt.js, Express.js**
for web portals and applications
- **Elastic Stack (Elasticsearch, Logstash, Kibana)**
for real-time analytics
- **Java Spring Boot and Golang microservices**
for backend processing
- **CI/CD pipelines and full containerization**
for efficient deployment cycles

Delivering tangible impact across stakeholders

Users praised the digital experience for its speed, transparency, and ease of use, particularly in comparison to previous manual procedures. Rental companies experienced faster approvals and better visibility, while customers enjoyed quicker, more secure booking.

Strategic benefits

Agile development: Implementing Agile and DevOps practices for faster, more reliable software delivery.

Microservices architecture: Using domain-driven design to build scalable and maintainable services.

API-first approach: Prioritizing API development to enable seamless integration and service accessibility.

Containerization: Utilizing Kubernetes and OpenShift for scalable and resilient deployments.

Blockchain: Enhancements for smart contracts and system upgrades for Hyperledger.

Elastic stack: Implementing ELK for analytics to enhance system functionality, reliability, and user engagement.

THIS' approach focused on leveraging technology to enhance transportation efficiency, customer satisfaction, and regulatory compliance.





Key elements include

Innovation & technology leadership: Embracing emerging technologies to drive digital transformation and set new benchmarks for efficiency and service modernization in the car rental ecosystem.

Enhanced customer experience: Delivering seamless, secure, and transparent digital services to improve convenience, trust, and satisfaction for end-users.

Robust compliance & oversight: Implementing rigorous monitoring mechanisms to uphold regulatory standards and ensure integrity across all rental operations.

Adaptability to Market Dynamics: Responding swiftly to evolving market demands and competitive landscapes through modular, scalable platform design.

Stakeholder-centric collaboration: Facilitating smooth integration and cooperation with ecosystem partners to ensure consistent and cohesive service delivery.

Key outcomes

Widespread reach

2,888 rental agencies
integrated, covering over
170,000 vehicles.

Revenue growth

**Platform-enabled
services** opened
new business channels.

Operational efficiency

Automation reduced

paperwork, errors,
and delays.

User satisfaction

**Secure and
user-friendly interface**
improved public trust.

Governance and oversight

Monitoring tools

enhanced regulatory control
without manual efforts

About Torry Harris

Torry Harris Integration Solutions (THIS) is a global leader in business, technology, and IT consulting services with over 25 years of expertise in digital transformation. Specializing in digital ecosystem enablement, marketplace implementation, full life-cycle API management, digital integration, AI, and Data, Torry Harris empowers enterprises to unlock new growth opportunities and accelerate innovation. Today, the company is helping businesses worldwide establish Global Capability Centers (GCCs) and drive transformative, AI-powered initiatives that deliver measurable impact.

Headquartered in New Jersey, USA, THIS operates multiple offshore development centers in Bangalore, India, and has a strong global footprint with offices in Bristol and Slough (UK), Dubai (UAE), Riyadh (KSA), Dublin (Ireland), Munich (Germany), Paris (France), and Valencia (Spain).



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